

EVEN SEMESTER EXAMINATION, 2023 – 24
2nd yr. B.Tech. – Computer Science & Engineering/Information Technology
Theory of Automata and Formal Languages

Duration: 3:00 Hrs.

Max Marks: 100

Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) Discuss Kleene Closure and languages in Automata. b) Explain productions with example. c) Discuss Arden's theorem with suitable example. d) Create Chomsky hierarchy of languages. e) How type 0 grammar is differing over type 1 grammar? f) Explain the different closure properties of Recursive languages.	5x4=20
Q 2.	Answer any four parts of the following. a) Differentiate between DFA and NFA. b) Construct a DFA and its transition table for the set of strings over the $\Sigma = \{0,1\}$ that ends in 101. c) Discuss the properties of regular languages. d) Explain pumping lemma for regular languages. e) Discuss minimization of finite automata. f) How Moore machine is differing over Mealy machine?	5x4=20
Q 3.	Answer any two parts of the following. a) Explain Pumping lemma for CFLs. Obtain a CFG for the language of palindrome over the alphabet $\{a, b, c\}$. b) Explain push down automata. Construct a PDA to accept a given language L by empty store where $L = \{a^n b^n \mid n \geq 1\}$ c) What is parse tree? Consider the grammar- $S \rightarrow A \mid B$ $A \rightarrow 0A \mid \epsilon$ $B \rightarrow 0B \mid 1B \mid \epsilon$ For the string $w=00101$, Identify the Leftmost derivation, Rightmost derivation and parse tree.	10x2= 20
Q 4.	Answer any two parts of the following. a) Explain Context-sensitive grammars and languages. Examine the grammar: $S \rightarrow A \mid B$ $A \rightarrow a A b \mid ab$ $B \rightarrow a b B \mid \epsilon$ Identify whether the grammar is ambiguous or not. b) Explain unrestricted grammars and equivalence with Turing machines. c) Explain Chomsky and Greibach normal forms. Find a grammar in Chomsky normal form equivalent to $S \rightarrow aAbB$, $A \rightarrow a A/a$, $B \rightarrow b B/b$	10x2= 20

Q 5.	Answer any two parts of the following. a) Explain about Church- Turing thesis and Universal Turing Machine. b) Explain Post Correspondence problem and unsolvable decision problem. c) What is Halting problem of TM? Design a Turing machine that erases all non-blank symbols on the tape, over the alphabets $\Sigma = \{a,b\}$	10x2= 20
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